

Unit I – Introduction to Design Thinking

Detailed lecture notes — what design thinking is, how it differs from ordinary problem solving, the five-stage process, human-centred design, and case studies of successful Indian and international innovations.

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SYLLABUS

UNIT I — INTRODUCTION TO DESIGN THINKING (6 periods)

Design thinking – concept and importance. Engineering design versus traditional problem solving. Innovation and creativity. Design thinking process – overview of the five stages. Human-centred design. Role of engineers in societal problem solving. Case studies of successful innovations.

Course Outcome CO1: Explain the principles and stages of the design thinking process.

Text: Tim Brown, *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation*, Harper Collins, 2012; Idris Mootee, *Design Thinking for Strategic Innovation*, 1st Edn., John Wiley & Sons, 2013.

References: Jeanne Liedtka and Tim Ogilvie, *Designing for Growth: A Design Thinking Toolkit for Managers*, Columbia Business School Publishing, 2011; Nigel Cross, *Engineering Design Methods: Strategies for Product Design*, 4th Edn., John Wiley & Sons, 2008.

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Learning objectives. After studying this unit you should be able to: (i) define design thinking and

explain why it matters to an engineer; (ii) distinguish a well-defined problem from an ill-defined or wicked one; (iii) contrast engineering design with traditional problem solving, and linear with iterative working; (iv) explain divergent and convergent thinking and the Double Diamond; (v) define creativity and innovation, list the barriers to creativity and classify innovation by technology and market; (vi) describe the five stages of the design thinking process and state what each stage produces; (vii) explain human-centred design in terms of desirability, feasibility and viability, and distinguish users, customers and stakeholders; and (viii) analyse a real innovation and identify the reframing on which it turned.

1. Introduction – what design thinking is about

Design thinking is a way of solving problems that begins with the people who will live with the solution rather than with the technology that is available to build it. It is both a **process** – a sequence of stages that a team can follow – and a **mindset**, a set of habits of thought that can be brought to any problem. Its output is not a drawing or a report; it is a solution that has been tested with real people and shown to work for them.

The method matters to engineers because the problems that arrive in professional practice are rarely as tidy as the problems set in a classroom. A practising engineer is far more often handed a situation than a question: costs are too high, a machine is not being used, a service is failing the people it was built for. Before anything can be calculated, somebody has to decide what the problem actually is – and that decision, far more than the calculation that follows, determines whether the work is worth doing at all.

1.1 A working definition

Design thinking is a human-centred, iterative approach to problem solving in which the problem itself is discovered and framed through direct contact with the people affected, and candidate solutions are made tangible and tested with those people before resources are committed.

Three phrases in that definition do the work. **Human-centred** means that the needs of real, specific people are the starting point and the final test, not an afterthought. **Iterative** means that the process loops: what is learned late sends the team back to a stage it has already passed. **Discovered and framed** means that the problem statement is an output of the work, not an input to it.

1.2 Why an engineering student should care

- **A technically perfect answer to the wrong question is worth nothing.** Accuracy in the wrong place is waste, however elegant the analysis.
- **Framing errors are the most expensive errors.** The cost of changing a decision rises steeply as a project advances; a mistake made while defining the problem is paid for again at every later stage.
- **Most products fail on desirability, not on engineering.** They work as designed and nobody wants them. No amount of further analysis detects this; only contact with users does.
- **It is a transferable method.** The same five stages apply to a product, a service, a public system or an organisational change.

KEY POINT Design thinking does not begin by asking “what can we build?” It begins by asking “whose problem is this, and what is it really?” Everything else in the method follows from that reversal.

1.3 What design thinking is not

Several misunderstandings are common enough to be worth naming. Design thinking is not styling or decoration added at the end of a project; the visual appearance of a product is one narrow part of design and not the part this course is about. It is not guesswork dressed up as creativity: every stage produces evidence that another person can examine. It is not a replacement for engineering analysis; analysis tells you whether a solution works, while design thinking tells you whether it is the right solution to build, and a competent engineer needs both. Finally, it is not a single brainstorming session held one afternoon – it is a cycle that is run many times.

Design thinking is not	Design thinking is
Decoration or styling added at the end	A discipline for finding the right problem
Guesswork presented as creativity	Evidence gathered from real users
A substitute for engineering analysis	A partner to analysis – it decides what to analyse
One brainstorm on one afternoon	A cycle run repeatedly until the evidence settles

2. Where the idea came from

Designers have worked in something like this way for as long as there have been designers. What is new is the deliberate description of that way of working as a method, and the teaching of it to people who are not designers – engineers, managers, doctors and public officials. The principal milestones are set out below.

Year	Milestone	What it contributed
1969	Herbert Simon, <i>The Sciences of the Artificial</i>	Argued that designing is a distinct mode of thinking, concerned with how things ought to be rather than how they are
1970s–80s	Design methods research; Nigel Cross	Showed by studying practising designers that they reason differently from scientists, working forward from tentative solutions
1991	IDEO formed	Made user observation and rapid prototyping an everyday professional practice rather than an occasional one
2005	Stanford d.school founded	Began teaching the process as a repeatable five-stage method to students of every discipline
2008	Tim Brown's article on design thinking	Brought the vocabulary into mainstream management and engineering practice

KEY POINT The way of working is old; the teaching of it as a repeatable process to non-designers is recent. You are not being asked to become a designer – you are being given a designer's method to use as an engineer.

3. Two kinds of problem – well-defined and wicked

The single most useful distinction in this unit is between problems that are fully specified when you receive them and problems that are not. The first kind is what your other courses set; the second kind is what professional practice sets.

3.1 Well-defined problems

A well-defined problem – sometimes called a **tame** problem – arrives complete. Every quantity needed is given and nothing that is not needed is given. There is one correct answer, and a marking scheme that already knows it. The method is taught before the problem is set, and the task is to execute that method accurately. “Calculate the deflection of a simply supported beam under a central point load” is such a problem. This is a necessary and valuable skill, and nothing in this course replaces it.

3.2 Ill-defined or wicked problems

An ill-defined problem arrives as a situation rather than a question. The problem itself has to be discovered. There are many workable answers and no single correct one; whether an answer is good depends on whose point of view is taken. Success is difficult to measure, and every solution changes the situation, so the problem you end with is not the problem you began with. The term **wicked problem** is used for the most extreme cases of this kind – problems in social planning, public health and infrastructure where the stakeholders disagree even about what would count as an improvement. “Make commuting safer for women” is such a problem.

Well-defined problem	Ill-defined (wicked) problem
■ Requirements are given ■ One correct answer exists ■ Success is easy to measure ■ Solve it once, it stays solved	■ The problem must be found first ■ Many answers, none 'correct' ■ Success depends on whose view ■ Each answer reshapes the problem
<i>“Calculate the beam deflection”</i>	<i>“Make commuting safer for women”</i>

Figure 1.1 The two kinds of problem. Engineering courses train the left-hand column; professional practice supplies the right.

Feature	Well-defined problem	Ill-defined (wicked) problem
The problem	Given, fully specified	Must be discovered and then framed
Information	Supplied with the question	Gathered from users in the field
Answers	One correct solution	Many workable options, none simply correct
Judging success	Against a known answer	Depends on whose view is taken
Stopping rule	When the answer is verified	When the result is good enough for now
Effect of solving	The problem is closed	Each solution reshapes the problem

KEY POINT Real problems do not announce which kind they are. The first act of design thinking is to

treat an unfamiliar situation as ill-defined until the evidence shows otherwise – the opposite of the classroom habit of assuming the question is complete.

4. Engineering design and traditional problem solving

4.1 The problems your other courses set

Traditional problem solving as taught in an engineering programme has a characteristic shape. The problem is given; the relevant theory has just been covered; the answer is single-valued; and failure is something to be avoided, because it is marked. This shape trains accuracy, method and perseverance, and it produces engineers who can be trusted with a calculation. What it does not train is the discovery of the question.

4.2 Linear and iterative working

The second difference is in the shape of the work itself. Traditional solving runs forward once: understand, analyse, compute, verify, report. Design thinking runs forward and back. A test at the end may send the team to the beginning; a conversation with a user during prototyping may show that the problem statement written three weeks earlier was wrong. These loops are not a sign that something has gone badly; they are the method working as intended.

Traditional linear problem solving



Design thinking – iterative

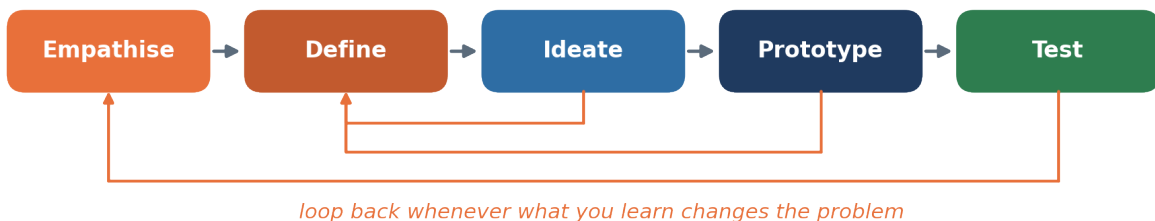


Figure 1.2 Linear and iterative working. The loops in the lower diagram are the characteristic feature of design thinking.

	Traditional problem solving	Design thinking
The problem	Given, fully specified	Discovered, then deliberately reframed
Information	Supplied with the question	Gathered from users in the field
The answer	One correct solution	Many workable options, one chosen
Failure	To be avoided	To be caused early, cheaply and on purpose
It ends when	The answer is verified	The user confirms that it works

Both columns are needed in practice. Most graduates arrive strong in the left-hand column and untrained in the right; the purpose of this course is to supply the right-hand column without weakening the left.

4.3 Analysis, synthesis and evaluation

Design uses three kinds of reasoning, and alternates between them. **Analysis** breaks a problem into parts and studies each – this is what the rest of an engineering curriculum trains. **Synthesis** puts parts together into a whole that did not previously exist; this is the characteristic move of the designer, and it is not reducible to analysis. **Evaluation** judges the whole against criteria and evidence, and sends the work back for change. A design process that contains only analysis has never produced a new object.

4.4 Problem-focused and solution-focused thinking

Nigel Cross, studying how expert designers actually work, found a pattern that distinguishes them from scientists. Scientists tend to work **problem-focused**: study the problem thoroughly and then derive the answer from that study. Designers tend to work **solution-focused**: propose a tentative solution early and use it as an instrument for understanding the problem. The proposal is not a commitment – it is a question put to the world. When it fails, the manner of its failure reveals what the problem really was.

KEY POINT A rough early solution is a research instrument, not a decision. This is why a team that sketches in week one usually understands the problem better by week three than a team that spends three weeks discussing it.

5. Divergent and convergent thinking; the Double Diamond

5.1 The two modes

Every design activity alternates between two modes of thinking. **Divergent thinking** opens a space: it generates as many options as possible, defers judgement, values quantity ahead of quality and ends with a wide field of possibilities. **Convergent thinking** closes that space: it compares, evaluates and eliminates, applies criteria and constraints, and ends with one option carried forward.

	Divergent thinking	Convergent thinking
Purpose	Generate options	Choose between options
Rule	Defer judgement entirely	Apply criteria deliberately
Measure	Quantity of ideas	Quality of judgement
Ends with	A wide field of possibilities	One option to carry forward
Typical tools	Brainstorming, mind mapping, SCAMPER	Dot voting, Pugh matrix, three-lens screening

KEY POINT Running both modes at the same time is what destroys good ideas. Separate them in time, and say aloud which mode the team is in. In a divergent session, “that will not work” is out of order – write it down and judge it later.

5.2 The Double Diamond

The Double Diamond is the standard picture of this alternation. It applies the divergent–convergent pair twice: once to the problem and once to the solution. In the first diamond the team explores the situation widely (**discover**) and then narrows to a single framed problem (**define**). In the second it generates many possible answers to that framed problem (**develop**) and narrows to one that is built and delivered (**deliver**).

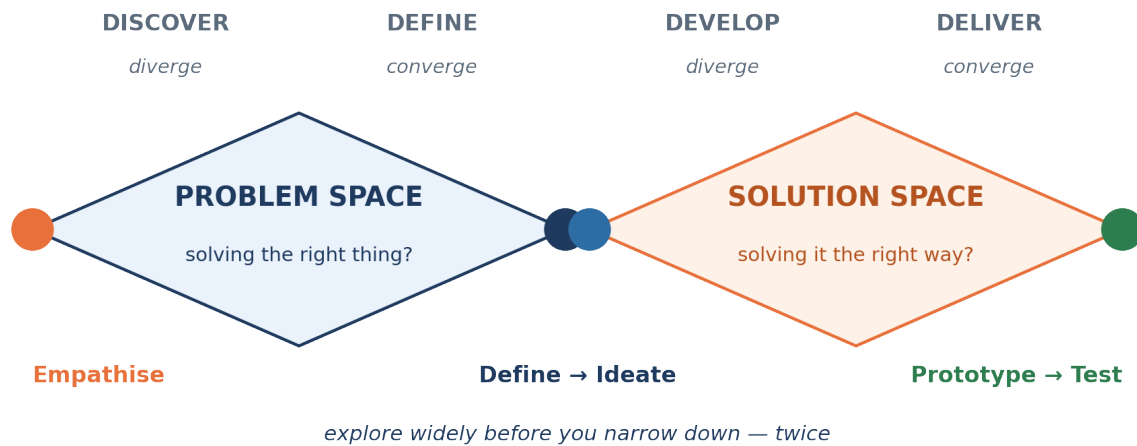


Figure 1.3 The Double Diamond. The narrow waist in the middle is the framed problem statement, the single most important output of the first half.

The most common error in student projects is to skip the first diamond entirely – to accept the problem as it was handed over and go straight to generating solutions. A team that does this may produce an excellent answer to a question nobody needed answered.

5.3 Why engineers find divergence difficult

- Years of training reward the fastest route to a single correct answer, so lingering among options feels like wasted time.
- Feasibility is judged instinctively, so weak-looking ideas are killed before anyone has had a chance to improve them.
- **Premature convergence** – the first workable idea quietly becomes the only idea on the table, and the team spends the remaining time defending it.
- Fear of appearing foolish in front of peers suppresses exactly the unusual suggestions that are worth having.

6. Innovation and creativity

6.1 What creativity means

Creativity is the production of something that is both **novel** and **useful**. Novelty alone is not creativity – an idea that is new and useless is merely unusual. Usefulness alone is not creativity either – an idea that is useful and familiar is routine. Creativity requires both at once.

Two further points matter for a student. First, creativity is a skill that improves with method and practice, not a fixed trait that some people are born with; the structured techniques in Unit III exist precisely because the output can be improved by procedure. Second, most creative output is **recombination** – the connecting of two things that were already present. This is why breadth of exposure matters more than raw inventiveness.

6.2 Measuring creative output

Measure	What it counts	Why it matters
Fluency	How many ideas were produced	The first measure, and the one most teams stop short on
Flexibility	How many different kinds of idea	Twenty variations of one idea score high on fluency and low here
Originality	How unusual an idea is compared with the rest of the group	Identifies the ideas nobody else would have offered

Measure	What it counts	Why it matters
Elaboration	How far an idea has been developed	A developed idea can be evaluated; a slogan cannot

6.3 Barriers to creativity

- **Perceptual** – seeing the problem in only one way, usually the way it was first described to you.
- **Emotional** – fear of being wrong, of being judged, or of appearing not to know.
- **Cultural** – “we have always done it like this”; the unstated rules of an organisation or a discipline.
- **Environmental** – no time, no space, constant interruption, or a senior person in the room whose opinion is known.
- **Intellectual** – reaching for the familiar method first, and framing every problem so that the familiar method applies.
- **Expressive** – having an idea but being unable to communicate it, so it never reaches anyone who could build on it.

6.4 Creative confidence

Creative confidence is the belief that you can change the things around you, together with the willingness to try. Most people have it as children and lose it during schooling, at the point where drawing and guessing begin to be marked. It is rebuilt through small successes: a rough sketch that somebody understood, a prototype that somebody used. Guided failure in a safe setting is the mechanism, and it is what the classroom activities in this course are for.

KEY POINT You do not need to be able to draw well. You need to be willing to be seen drawing badly. Almost every barrier in the list above is a social barrier rather than a cognitive one.

7. Invention, innovation and the four types of innovation

7.1 Invention and innovation

The two words are often used interchangeably and should not be. An **invention** is a new idea, device or method, proven in a laboratory or a workshop and measured by novelty and by patents; an invention can sit unused for decades. An **innovation** is an invention put into use at scale, proven in the market or in the field and measured by adoption and impact; it requires a business model, a supply chain and a user.

Basis	Invention	Innovation
What it is	A new idea, device or method	An invention put into real use
Proven where	In the laboratory or workshop	In the market or in the field
Measured by	Novelty; patents granted	Adoption, scale and impact
Also needs	Technical skill	A model, a supply chain and a user
Failure mode	Never leaves the bench	Reaches nobody, or cannot be sustained

KEY POINT Invention costs money; innovation is what makes the invention matter. A student project that works on the bench and reaches nobody has produced an invention, not an innovation.

7.2 Four types of innovation

Innovation is usefully classified by two questions: is the technology new, and is the market new? The four combinations are shown below.

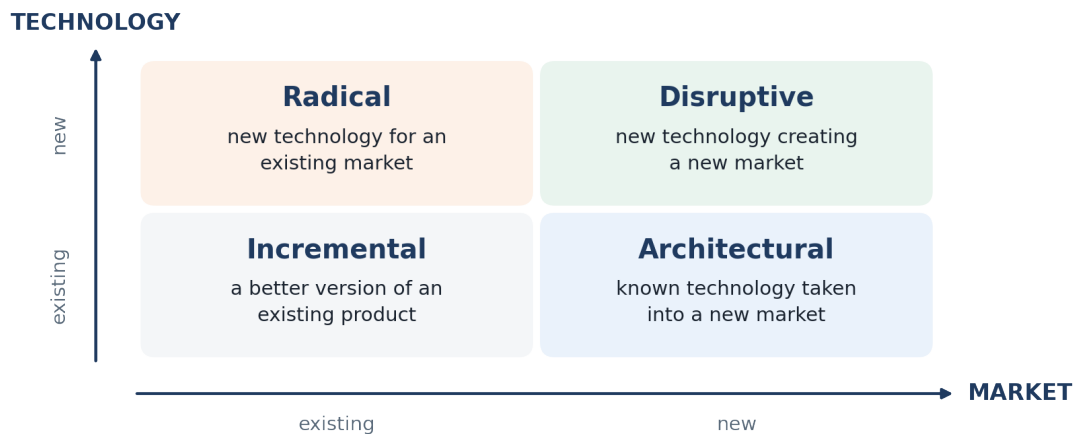


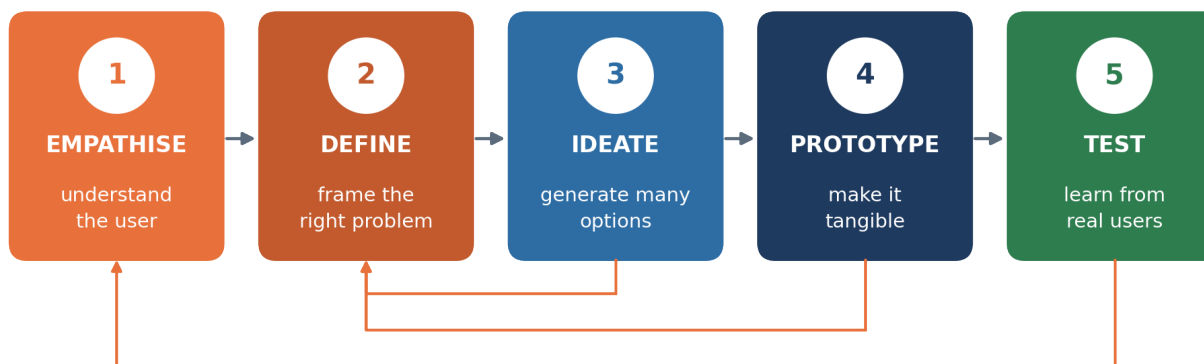
Figure 1.4 Innovation classified by technology and market. Design thinking applies in all four quadrants, not only in the top-right one.

Type	What is new	Example
Incremental	Neither	Each successive model of a domestic mixer grinder
Architectural	The market	A clay cooling pot sold as a household refrigerator
Radical	The technology	LED lamps replacing incandescent filament bulbs
Disruptive	Both	A portable, mains-free infant warmer for home births

Most professional engineering work is incremental, and design thinking is just as useful there as at the frontier: it is how an existing product stays worth buying. The method is not reserved for breakthroughs.

8. The design thinking process – the five stages

The model used in this course is the five-stage model taught at the Stanford d.school. The five stages are not a timetable to be marched through once; they are a checklist of moves to be made, in any order the evidence demands, and usually more than once.



the loops matter as much as the stages

Figure 1.5 The five-stage design thinking process. The loops beneath the stages matter as much as the stages themselves.

8.1 Stage 1 – Empathise

The purpose of the first stage is to set aside what you assume and find out how the user actually experiences the situation. The work is done where the problem lives: you observe people in their own setting and you interview them about specific past events rather than about their opinions. The critical discipline is to look at what people **do** and not only at what they **say** they do, because habits are invisible to the person who has them. **Produces:** field notes, photographs, verbatim quotations and empathy maps.

8.2 Stage 2 – Define

The second stage turns everything gathered into one clear, actionable problem statement. A good statement names the user specifically, states the need as something the user must be able to **do**, and gives the insight that explains why the need exists and is not already met. It is narrow enough to act on and wide enough to allow many different answers, and it contains no solution. **Produces:** a point-of-view statement and a set of “How might we...” questions.

KEY POINT Define is the stage engineers most often skip, and it is the stage that decides whether everything after it was worth doing. A team that cannot write its problem in one sentence does not yet have a problem – it has a topic.

8.3 Stage 3 – Ideate

The third stage generates a wide field of options before any one of them is chosen. Structured techniques – brainstorming, brainwriting, mind mapping, SCAMPER, lateral thinking – are used to push past the obvious answers, and convergence is then carried out deliberately, against criteria agreed in advance. **Produces:** a large set of ideas, of which two or three are chosen to build.

8.4 Stage 4 – Prototype

The fourth stage makes the idea tangible as quickly and as cheaply as possible. Paper, cardboard, clay, a role play or a sequence of hand-drawn screens are all legitimate prototypes; fidelity is kept low on purpose, because a rough object invites criticism while a polished one invites politeness. Each prototype should be built to answer one specific question. **Produces:** something a user can hold, use or walk through.

8.5 Stage 5 – Test

The fifth stage puts the prototype in front of real users and watches what they do with it. The user is asked to use it, not to judge it; the team says as little as possible, because every intervention deletes a finding. What is recorded is what surprised the team, because surprises are where the learning is. **Produces:** evidence that usually sends the team back to an earlier stage with better information.

Testing rarely ends a project. It normally sends the team back to Define or Ideate – and that return journey is the method working, not failing.

9. Other models of the same process

9.1 IDEO's three spaces

Tim Brown describes the work not as a sequence of steps but as three overlapping **spaces** that a project moves between, often more than once: **inspiration**, the problem or opportunity that starts the search; **ideation**, the generating, developing and testing of ideas; and **implementation**, the path from the project room into people's lives.

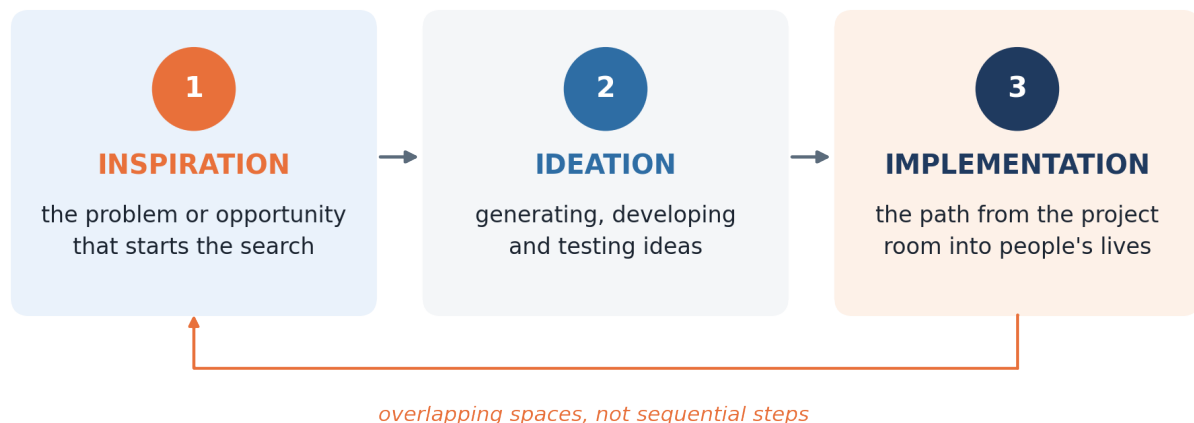


Figure 1.6 IDEO's three spaces of innovation. They overlap and projects loop back through them repeatedly.

9.2 The Design Council's four Ds

The British Design Council describes the same territory as Discover, Define, Develop and Deliver – the four quadrants of the Double Diamond met in Section 5. The three models are compared below.

d.school – five stages	IDEO – three spaces	Design Council – four Ds
Empathise	Inspiration	Discover
Define	Inspiration	Define
Ideate	Ideation	Develop
Prototype	Ideation	Develop
Test	Implementation	Deliver

9.3 What every model agrees on

- Understand the people before deciding what to build.
- Frame the problem explicitly, and in writing.
- Open the solution space wide before closing it.
- Learn by building something rough rather than by arguing about it.
- Expect to go round again – the loop is the method, not a symptom of failure.

KEY POINT The models differ in vocabulary and in the number of boxes drawn. They do not differ in the moves they ask you to make. If a process model has no loop in it, it is not describing design.

10. Human-centred design – desirability, feasibility and viability

10.1 What human-centred means

Human-centred design means that the people affected are the starting point of the design and not a test applied at the end. Their needs are found by observing and listening rather than assumed from the designer's own experience; they take part throughout, in framing the problem, in choosing between options and in testing; and success is judged by what changes for them rather than by what the team managed to build.

KEY POINT You are almost never the user. Designing for yourself is the commonest and the costliest mistake in student projects, because your age, education, income and confidence are all unusual by comparison with the population you are designing for.

10.2 The three lenses

Any idea that is to become an innovation must survive three tests. **Desirability** asks whether people actually want it. **Feasibility** asks whether it can be built with the technology and skills available. **Viability** asks whether it can be sustained – financially, operationally and organisationally. Innovation lies where all three overlap.

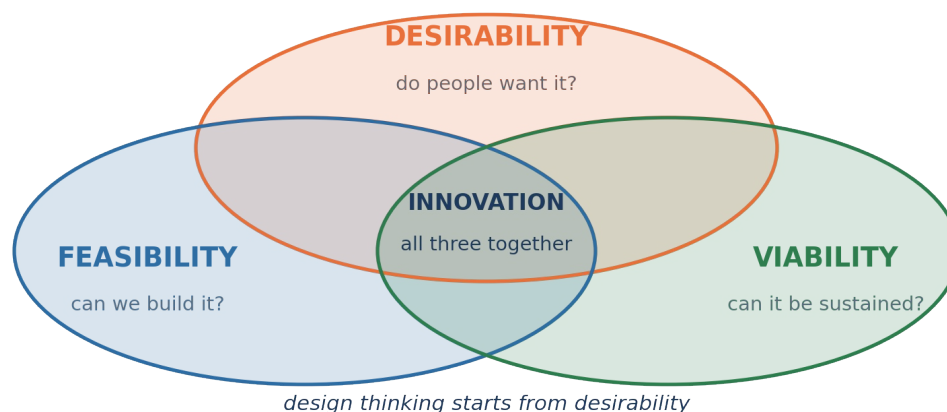


Figure 1.7 The three lenses of innovation. Design thinking approaches the overlap starting from desirability.

What is missing	What you get	Typical symptom
Not viable	People love it and it can be built, but it cannot be sustained	A pilot that works beautifully and dies when the grant ends
Not feasible	People want it and it would pay, but it cannot be built	A promise that depends on technology that does not exist yet
Not desirable	It can be built and it would pay, but nobody wants it	A well-engineered, profitable product that sits unsold

KEY POINT Design thinking starts from desirability because feasibility and viability can be engineered afterwards, while desirability cannot be added at the end. A product nobody wants cannot be made wanted by improving its specification.

11. Users, customers and stakeholders

A recurring source of failure is the assumption that the person who uses a product is the same person who pays for it and the same person who can approve it. In practice these are frequently three different people with three different sets of needs, and a design that satisfies only one of them does not get adopted.

Role	What they do	Example in a campus water system
User	Uses the thing, and lives with every design decision	The student who fills a bottle between classes
Customer	Pays for it; often never touches it	The institution's estates department
Gatekeeper	Can permit or forbid its use	The health and safety officer; the maintenance contractor
Influencer	Shapes the decision without making it	A senior faculty member; the existing supplier

KEY POINT Ask three questions at the start of every project: who uses it, who pays for it, and who can say no to it. Projects fail far more often on a stakeholder nobody mapped than on a technical problem nobody solved.

12. Inclusive design and extreme users

If a design is made for the average user, the people at the edges of the range are excluded. If it is made for the edges, the middle is almost always served as well. This asymmetry is the argument for inclusive design, and it is a practical argument rather than a moral one – although it is also a moral one.

Extreme users – the expert who pushes a product beyond its intended use and the novice who cannot begin to use it – meet every weakness in a design sooner and harder than average users do. They have usually already invented workarounds, and a workaround is free design research for anyone willing to look at it. Interviewing three extreme users commonly teaches more, and teaches it faster, than interviewing thirty average ones.

Ability is also not fixed. It changes with age, injury, tiredness, noise and light, and many exclusions are temporary or situational rather than permanent: a broken arm, a crowded bus, bright sunlight on a screen, a language that is not your first. The four exclusions that engineers most often overlook are cost, literacy and language, physical ability, and the assumption of reliable electricity and connectivity.

KEY POINT A ramp built for a wheelchair carries every trolley, pram and delivery cart on the campus. Accessibility is not a special case bolted on afterwards; it is ordinary good design that somebody bothered to check.

13. The role of engineers in societal problem solving

13.1 Engineering is a social activity

Every artefact an engineer designs changes how somebody lives, works or travels. Technical choices carry social consequences that no equation reports: who is included, who is excluded and who bears the cost are design decisions, whether or not anyone notices making them. India's largest problems – safe

water, affordable health care, energy, mobility, sanitation – are engineering problems with people at their centre, and they are not solved by technology alone.

13.2 Where engineers meet society

Domain	The engineering problem	The human problem inside it
Water and sanitation	Treatment, storage and distribution	Households with no reliable supply, no electricity and no plumber
Health and care	Devices, diagnostics and clinic equipment	Staff, power and transport that are not what the device assumed
Energy and climate	Efficiency, renewable generation, cooling	Running cost that a household can actually afford month by month
Mobility and safety	Transport systems and public space	Usability by women, children, older people and the disabled

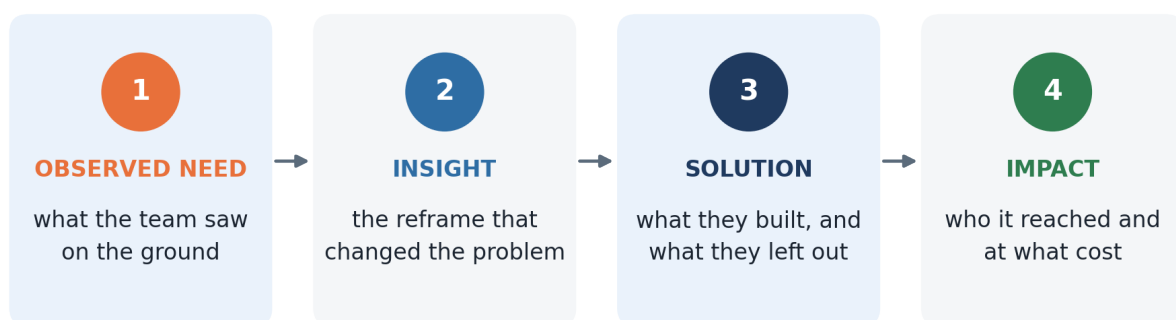
13.3 Professional and ethical responsibility

The Institution of Engineers (India) asks its members to use their knowledge for the welfare, health and safety of the community without discrimination; to act only within their competence and with care, sincerity and honesty; to be fair and honest in public statements on the basis of adequate knowledge; and to inform employers and clients of the environmental, economic and social consequences of their work. That last obligation is the one most relevant to this unit: the consequences of a design are part of the design, and reporting them is not optional.

KEY POINT If you have not decided whom your design serves, you have still decided – you simply did not watch yourself do it. “It was technically correct” has never been accepted as a defence.

14. Case studies of successful innovations

The five cases below are the core of this unit. Each is read in four moves: the need that was observed on the ground, the insight or reframing that changed the problem, the solution that was built, and the impact it achieved. In every case the technology is modest and the reframing is what was clever.



the insight is where the design thinking happened

Figure 1.8 How to read a design case. The second move – the insight – is where the design thinking happened.

CASE STUDY 1.1 — Embrace Infant Warmer – the brief that turned out to be wrong

Observed need. A class at the Stanford d.school was set a brief in its Entrepreneurial Design for Extreme Affordability course: design an infant incubator costing one per cent of a conventional one,

which in the United States cost about twenty thousand dollars. Low birth weight and hypothermia kill very large numbers of newborns in low-income countries, and incubators were assumed to be the answer.

Insight. In 2007 a member of the team, Linus Liang, travelled to Nepal and found that the brief was wrong. Most at-risk babies were not born in hospitals at all; they were born in villages, far from any clinic, where electricity is unreliable and trained staff are few. A cheaper incubator standing in a city hospital would never reach them. The real problem was not the cost of an incubator – it was keeping a newborn warm without mains power, in a home, in the hands of the mother herself.

Solution. A small sleeping bag containing a removable pouch of phase-change wax. The pouch is warmed in hot water and then holds a temperature of about 37 °C for roughly four hours. It needs no continuous electricity and no technician, and a mother can use it herself. Material cost fell from the order of twenty thousand dollars to the order of twenty-five.

What to learn from it. The team's most valuable finding was not a design. It was the discovery that the problem they had been given was the wrong problem. No amount of engineering skill applied to the original brief would have produced this result.

CASE STUDY 1.2 — Aravind Eye Care System – designing the business model

Observed need. Dr Govindappa Venkataswamy founded Aravind in Madurai in 1976 with eleven beds. Cataract blindness was, and remains, largely curable; the obstacle was not surgical technique but cost, distance and sheer capacity.

Insight. If the surgery itself is not the problem, then the design object is the system that delivers it. Dr V studied how a fast-food chain achieved consistency at very high volume, and applied the same thinking to eye surgery: standardised procedures, an assembly-line theatre in which a surgeon moves between prepared tables, and a cross-subsidy under which paying patients fund the roughly half of all patients who pay little or nothing.

A second reframing. Imported intraocular lenses cost hundreds of dollars each and capped the number of people who could be treated. Rather than negotiate the price, Aravind established its own manufacturing arm, Aurolab, in 1992. Lens cost fell to a few dollars and the total cost of a cataract operation fell to approximately ten.

What to learn from it. When a constraint blocks the mission, design thinking treats the constraint itself as the next problem to be designed away. The system now performs hundreds of thousands of surgeries a year with complication rates comparable to the best in the world.

CASE STUDY 1.3 — The Jaipur Foot – designed for the life the user actually lives

Observed need. Prosthetic feet available in India in the 1960s were imported designs. They were expensive, and they were built on assumptions about daily life that did not hold for most Indian users.

Insight. Dr P. K. Sethi, an orthopaedic surgeon at Sawai Man Singh Medical College, working with Ram Chandra Sharma, a craftsman, identified the mismatch precisely: existing prosthetic feet did not allow the wearer to go barefoot, to squat, to sit cross-legged or to work in wet fields. These are not optional activities in rural India; they are how ordinary life is conducted.

Solution. The Jaipur Foot, developed in 1968, uses a rubber core, a wooden block and a vulcanised rubber coating – later versions use polyurethane. It permits squatting, cross-legged sitting and barefoot walking on uneven ground, can be made and fitted locally in a day, and costs a small fraction of a Western prosthesis.

What to learn from it. The imported foot was not badly engineered. It was engineered for a life its users did not live. Bhagwan Mahaveer Viklang Sahayata Samiti has fitted it free of charge to well over two million people.

CASE STUDY 1.4 — MittiCool – a constraint treated as the design brief

Observed need. Mansukhbhai Prajapati, from a potter's family in Morbi district in Gujarat, had worked with clay since 1989. Households without a reliable electricity supply had no way at all to keep food from spoiling, and a conventional refrigerator was out of reach on both counts – purchase price and running supply.

Insight. The absence of electricity was treated as a constraint to design within rather than a problem to be solved first. Evaporative cooling in porous clay is an old principle; what was new was applying it in the form of a household appliance with shelves, a door and a water tap.

Solution. A two-chamber clay cabinet whose walls hold water; evaporation keeps the interior roughly eight degrees below room temperature. Vegetables keep for about four days and milk for about two. It needs no electricity, is repairable, and is sold at a price of the order of three thousand rupees.

What to learn from it. The innovation here is architectural rather than technological: a known technology was taken into a market that had never been served by it. The enterprise was built by the innovator himself.

CASE STUDY 1.5 — Tata Swach – a waste material becomes the product

Observed need. Relief filters deployed by Tata Chemicals after the Indian Ocean tsunami of 2004 exposed what a household water filter has to survive in practice: no mains electricity, no running water at pressure, no plumber, and a purchase price and running cost that a low-income household can meet.

Insight. Two reframings mattered. The filter medium did not have to be an expensive engineered material – rice husk ash, an agricultural waste product, could carry nano-silver particles that kill bacteria. And the product's economics should be stated as a monthly running cost rather than a purchase price, because that is how the household actually decides.

Solution. Tata Swach was launched in December 2009. It is gravity-fed, so it needs neither electricity nor running water. The cartridge purifies about three thousand litres and then stops the flow altogether, so it cannot be used unsafely past its life. The running cost was of the order of thirty rupees a month for a family of five.

What to learn from it. The fail-safe cartridge is the detail worth noticing. It is a design response to an observation about behaviour – that people do not replace filters on schedule – rather than to a specification.

14.1 What the five cases have in common

Case	The reframing	What made it work
Embrace	Villages, not hospitals	Phase-change wax pouch; no mains power
Aravind	Volume and cost, not technique	Cross-subsidy and its own lens factory
Jaipur Foot	A barefoot life, not a shod one	Rubber and wood, fitted in a day
MittiCool	No electricity at all	Evaporative cooling in clay
Tata Swach	No power and no plumbing	Rice husk ash with nano-silver

- In every case somebody went and looked, and came back with a different problem from the one they were sent to solve.
- The binding constraint – cost, power, distance, literacy – became the design brief instead of an excuse.

- The technology is modest in all five; the framing is what was clever.
- Each was designed for how people actually live rather than for how they ought to live.
- All five reached scale, which is what separates an innovation from an invention.

KEY POINT In every one of these cases the decisive work happened before any engineering began. That is the claim this unit is making, and the five cases are the evidence for it.

IN CLASS Take one of the five cases, or bring another. In one line each, write the observed need, the insight, the solution and the impact. Then state what the brief would have been if nobody had gone into the field. Present in three minutes; the class judges whether the reframing is real.

15. Summary

- Design thinking is a human-centred, iterative approach in which the problem is discovered and framed through contact with the people affected, and solutions are made tangible and tested before resources are committed.
- Well-defined problems are given, closed and singly answered; ill-defined or wicked problems must first be discovered, admit many answers and are reshaped by every attempt to solve them.
- Traditional problem solving runs forward once; design thinking loops, and the loops are the method rather than a symptom of failure.
- Design alternates analysis, synthesis and evaluation; designers reason solution-focused, using a rough early proposal as an instrument for understanding the problem.
- Divergence opens a space and convergence closes it, never at the same time; the Double Diamond applies that pair twice – once to the problem and once to the solution.
- Creativity is novelty plus usefulness; it is measured by fluency, flexibility, originality and elaboration, and it is blocked by perceptual, emotional, cultural, environmental, intellectual and expressive barriers.
- An invention becomes an innovation only when it is put into use at scale; innovation is classified as incremental, architectural, radical or disruptive by whether the technology and the market are new.
- The five stages are Empathise, Define, Ideate, Prototype and Test; each produces something specific that the next stage consumes, and Define is the stage most often skipped and the one that decides the outcome.
- IDEO's three spaces and the Design Council's four Ds describe the same moves under different names; every model of the process contains a loop.
- An idea must clear desirability, feasibility and viability; design thinking begins at desirability because the other two can be engineered afterwards and desirability cannot.
- User, customer, gatekeeper and influencer are four different roles; designing for the edges of the range serves the middle as well.
- In all five case studies the technology was modest and the reframing was decisive; the binding constraint became the design brief.

16. Review questions

16.1 Short-answer (two-mark) questions

1. Define design thinking in one sentence.
2. State two differences between a well-defined and an ill-defined problem.

3. What is meant by a wicked problem? Give one example.
4. Distinguish divergent thinking from convergent thinking.
5. What are the four quadrants of the Double Diamond?
6. Define creativity. Why is novelty alone insufficient?
7. Name the four measures of creative output.
8. List any four barriers to creativity.
9. Differentiate between invention and innovation with one example of each.
10. Classify innovation by technology and market, naming all four types.
11. Name the five stages of the design thinking process in order.
12. What does the Empathise stage produce?
13. Why is the Define stage considered the most critical?
14. State the three lenses of innovation.
15. Distinguish a user from a customer and from a gatekeeper.
16. What is an extreme user, and why interview one?
17. State any two obligations in the Institution of Engineers (India) code of ethics.
18. What was the reframing in the Embrace Infant Warmer case?
19. What is meant by cross-subsidy, with reference to Aravind Eye Care?
20. Why does the Tata Swach cartridge stop the flow after about three thousand litres?

16.2 Long-answer (thirteen-mark) questions

1. Explain design thinking as both a process and a mindset. Discuss why it is relevant to an engineering graduate, with reference to the cost of framing errors.
2. Compare well-defined and ill-defined problems under at least six headings. Illustrate each with an engineering example and explain why classroom training prepares students for only one of them.
3. Contrast engineering design with traditional problem solving. Discuss linear and iterative working, the roles of analysis, synthesis and evaluation, and Nigel Cross's distinction between problem-focused and solution-focused reasoning.
4. Explain divergent and convergent thinking and describe the Double Diamond model. Discuss why engineers commonly find divergence difficult and how a team can guard against premature convergence.
5. Define creativity and innovation, and distinguish them from invention. Describe the four measures of creative output and the six barriers to creativity, with examples drawn from student project work.
6. Describe the five stages of the design thinking process. For each stage state its purpose, the principal activities and the output it produces, and explain why the process is drawn with loops.
7. Compare the d.school five-stage model, IDEO's three spaces and the Design Council's four Ds. Identify the moves on which all three agree and explain why every model contains a loop.
8. Explain human-centred design in terms of desirability, feasibility and viability. Discuss what happens when each lens in turn is missing, and explain why design thinking begins from desirability.
9. Discuss the distinction between users, customers, gatekeepers and influencers with an example from a campus or municipal service. Explain how a failure to identify all four can cause a technically sound project to fail.
10. Explain inclusive design and the role of extreme users. Discuss the four exclusions engineers most often overlook, with an example of each from Indian practice.
11. Discuss the role of the engineer in societal problem solving with reference to the Institution of Engineers (India) code of ethics. Explain why the social consequences of a design are part of the design.

12. Describe any three of the case studies in this unit. In each case identify the observed need, the insight or reframing, the solution and the impact, and state what the brief would have been without field work.
13. “In every successful innovation studied in this unit, the decisive work happened before any engineering began.” Discuss this statement critically, using evidence from at least three cases, and state the limits of the claim.

Text: Tim Brown, *Change by Design*, Harper Collins, 2012 — Chapters 1–4; Idris Mootee, *Design Thinking for Strategic Innovation*, Wiley, 2013 — Chapters 1–2. **References:** Jeanne Liedtka and Tim Ogilvie, *Designing for Growth*, Columbia Business School Publishing, 2011; Nigel Cross, *Engineering Design Methods*, 4th Edn., Wiley, 2008 — Chapters 1 and 2; Herbert A. Simon, *The Sciences of the Artificial*, MIT Press, 1969. **Case sources:** Stanford Magazine and Embrace Innovations (Embrace); Aravind Eye Care System and Aurolab (Aravind); Bhagwan Mahaveer Viklang Sahayata Samiti (Jaipur Foot); MittiCool and the National Innovation Foundation (MittiCool); Tata Chemicals press release, 7 December 2009 (Tata Swach).
